

Problem Set 1

Due: September 27, 2026, at 11:59 p.m.

1. [20 points] Consider a venture that succeeds with probability $1/2$. For each dollar invested, you earn a profit of \$1.60 if the venture succeeds and lose the invested dollar if it fails. You may invest repeatedly, and each time you invest exactly half of your current wealth. Starting with \$1,000,000:

- (i) **[5 points]** What is the effect of combining this investment opportunity with the strategy of always investing half of your current wealth?
- (ii) **[5 points]** Do you gain or lose money on average?
- (iii) **[10 points]** If you make 10,000 independent investments of this kind, what is your expected wealth after the final investment? What is the probability that your final wealth is less than the \$1,000,000 with which you started?

Solution.

(i) Denote the wealth after n investments by W_n . Denote the outcome of the n -th investment by $X_n := \mathbb{1}\{\text{success}\}$, which is a Bernoulli random variable with success probability $1/2$. Then

$$W_n = \begin{cases} (1 + 1.6) \frac{W_{n-1}}{2} + \frac{W_{n-1}}{2} = 1.8W_{n-1}, & \text{if } X_n = 1, \\ 0 \times \frac{W_{n-1}}{2} + \frac{W_{n-1}}{2} = 0.5W_{n-1}, & \text{if } X_n = 0. \end{cases}$$

(ii) Given last step's wealth W_{n-1} , the expected wealth after the n -th investment is

$$\mathbb{E}[W_n | W_{n-1}] = 1.8 \cdot \frac{1}{2} W_{n-1} + 0.5 \cdot \frac{1}{2} W_{n-1} = 1.15W_{n-1} > W_{n-1}.$$

Therefore, on average, you gain money with this strategy.

(iii) The expected wealth after n investments is

$$\mathbb{E}[W_n] = \mathbb{E}[\mathbb{E}[W_n | W_{n-1}]] = 1.15\mathbb{E}[W_{n-1}] = 1.15^n W_0.$$

Therefore, after 10,000 investments, the expected wealth is

$$\mathbb{E}[W_{10000}] = 1.15^{10000} \cdot 10^6.$$

On the other hand, denote the total number of successful investments by $S_n = \sum_{i=1}^n X_i$. Then total wealth after n investments is

$$W_n = 10^6 \times 1.8^{S_n} \times 0.5^{n-S_n}$$

To have $W_n < 10^6$, we need $S_n < n \frac{\log 2}{\log 3.6}$. Substituting $n = 10000$ gives $S_{10000} < 10000 \frac{\log 2}{\log 3.6} \approx 5411.264$, hence $W_{10000} < 10^6 \iff S_{10000} \leq 5411$. Since $S_n \sim \text{Binomial}(10000, 1/2) \approx \mathcal{N}(5000, 2500)$, we can use the normal approximation to estimate the probability:

$$\mathbb{P}(W_{10000} < 10^6) = \mathbb{P}(S_{10000} \leq 5411) \approx \mathbb{P}\left(Z \leq \frac{5411 - 5000}{\sqrt{10000 \cdot 1/2 \cdot 1/2}}\right) = \mathbb{P}(Z \leq 8.22) \approx 1.$$

2. [10 points] A standard 52-card deck is well shuffled, and five cards are dealt without replacement. Find the probability of each of the following mutually exclusive poker-hand categories: royal flush; other straight flush; four of a kind; full house; flush (excluding straight flushes); straight (excluding straight flushes); three of a kind; two pairs; one pair; and high card only (none of the preceding categories). Show how you count the number of hands in each category, and verify that your counts sum to $\binom{52}{5}$.

Solution.

The total number of five-card hands is

$$\binom{52}{5} = 2,598,960.$$

- **Royal flush:** 10, J, Q, K, A of the same suit.

$$\frac{4}{\binom{52}{5}} = \frac{4}{2,598,960} \approx 0.000001539077.$$

- **Other straight flush:** five consecutive ranks of the same suit, excluding royal flushes.

$$\frac{10 \cdot 4 - 4}{\binom{52}{5}} = \frac{36}{2,598,960} \approx 0.000013851695.$$

- **Four of a kind:** four cards of one rank and one card of another rank.

$$\frac{13 \cdot 48}{\binom{52}{5}} = \frac{624}{2,598,960} \approx 0.000240096038.$$

- **Full house:** three cards of one rank and two cards of another rank.

$$\frac{13 \binom{4}{3} \cdot 12 \binom{4}{2}}{\binom{52}{5}} = \frac{3744}{2,598,960} \approx 0.001440576230.$$

- **Flush:** five cards of the same suit, excluding straight flushes.

$$\frac{4 \binom{13}{5} - 40}{\binom{52}{5}} = \frac{5108}{2,598,960} \approx 0.001965401545.$$

- **Straight:** five consecutive ranks, excluding straight flushes.

$$\frac{10 \cdot 4^5 - 40}{\binom{52}{5}} = \frac{10200}{2,598,960} \approx 0.003924646782.$$

- **Three of a kind:** three cards of one rank and two cards of two distinct other ranks.

$$\frac{13 \binom{4}{3} \binom{12}{2} 4^2}{\binom{52}{5}} = \frac{54912}{2,598,960} \approx 0.021128451381.$$

- **Two pairs:** two cards of one rank, two cards of another rank, and one card of a third rank.

$$\frac{\binom{13}{2}\binom{4}{2}^2 \cdot 11 \cdot 4}{\binom{52}{5}} = \frac{123552}{2,598,960} \approx 0.047539015606.$$

- **One pair:** two cards of one rank and three cards of three distinct other ranks.

$$\frac{13\binom{4}{2}\binom{12}{3}4^3}{\binom{52}{5}} = \frac{1098240}{2,598,960} \approx 0.422569027611.$$

- **High card only:** five distinct ranks, not consecutive, and not all of the same suit.

$$\frac{\left(\binom{13}{5} - 10\right)(4^5 - 4)}{\binom{52}{5}} = \frac{1302540}{2,598,960} \approx 0.501177394035.$$

Finally,

$$4 + 36 + 624 + 3744 + 5108 + 10200 + 54912 + 123552 + 1098240 + 1302540 = 2,598,960 = \binom{52}{5}.$$

3. [10 points] Let $\mathcal{F}$ be a σ -field on Ω . Prove that $\mathcal{F}$ is closed under intersections and set differences; that is, for any $A, B \in \mathcal{F}$, show that

$$A \cap B \in \mathcal{F} \quad \text{and} \quad A \setminus B \in \mathcal{F}.$$

Solution.

Since $A \cap B = (A^c \cup B^c)^c$, and $\mathcal{F}$ is closed under complements and countable unions, we have $A^c \cup B^c \in \mathcal{F}$, and thus $A \cap B \in \mathcal{F}$.

Also, since $A \setminus B = A \cap B^c$, and we have already shown that $\mathcal{F}$ is closed under intersections and complements, we conclude that $A \setminus B \in \mathcal{F}$.

4. [10 points] Suppose that we are given a measure space $(\Omega, \mathcal{F}, \mu)$. Show that whenever $A \subseteq B$, we have

$$\mu(A) \leq \mu(B).$$

Make explicit at each step where the defining properties of a measure and a σ -field are used.

Solution.

If $A = B$, then $\mu(A) = \mu(B)$, and the inequality holds.

If $A \subsetneq B$, then $B \setminus A \in \mathcal{F}$ and is nonempty. This is because $A^c \in \mathcal{F}$ by the closure under complements of σ -fields, and then $(B \setminus A) = B \cap A^c \in \mathcal{F}$ by the closure under intersections of σ -fields. Since A and $B \setminus A$ are disjoint measurable sets, countable additivity of μ gives

$$\mu(B) = \mu(A \uplus (B \setminus A)) = \mu(A) + \mu(B \setminus A).$$

By nonnegativity of measures, $\mu(B \setminus A) \geq 0$, and hence $\mu(B) \geq \mu(A)$.

5. [20 points] Let $\Omega = \{R, M, L\}$ and take the σ -field $\mathcal{F} = 2^\Omega$. Let P be a probability measure on $(\Omega, \mathcal{F})$ such that

$$P(\{R\}) = \frac{1}{2}, \quad P(\{M\}) = \frac{1}{4}, \quad P(\{L\}) = \frac{1}{4}.$$

Define random variables X and Y by, for every $\omega \in \Omega$,

$$X(\omega) = 2 \mathbf{1}\{\omega = R\} + 3 \mathbf{1}\{\omega \in \{M, L\}\},$$

$$Y(\omega) = 2 \mathbf{1}\{\omega \in \{M, L\}\} + 3 \mathbf{1}\{\omega = R\}.$$

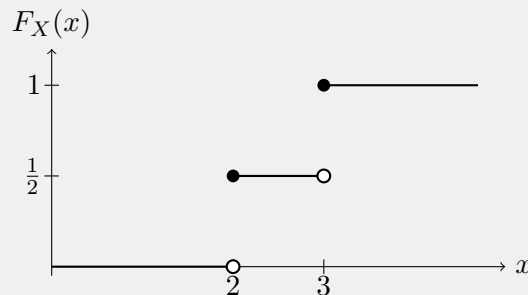
- (i) [10 points] Find the cumulative distribution function of X and sketch its graph.
- (ii) [10 points] Find $\sigma(X)$, the σ -field generated by X .

Solution.

(i) $\mathbb{P}(X = 2) = \mathbb{P}(\{R\}) = 1/2$, $\mathbb{P}(X = 3) = \mathbb{P}(\{M, L\}) = 1/4 + 1/4 = 1/2$. Therefore, the cumulative distribution function of X is

$$F_X(x) = \begin{cases} 0, & x < 2, \\ 1/2, & 2 \leq x < 3, \\ 1, & x \geq 3. \end{cases}$$

The graph of $F_X(x)$ is as follows.



(ii) The σ -field generated by X is

$$\sigma(X) = \{\emptyset, \{R\}, \{M, L\}, \Omega\}.$$

6. [20 points]

Suppose that we are given a measurable space $(\Omega, \mathcal{F})$.

- (i) [10 points] Suppose that $f : \Omega \rightarrow \mathbb{R}^d$ is measurable and $g : \mathbb{R}^d \rightarrow \mathbb{R}^r$ is Borel measurable; that is, for every $B \in \mathcal{B}(\mathbb{R}^r)$,

$$g^{-1}(B) \in \mathcal{B}(\mathbb{R}^d).$$

Show that $g \circ f$ is measurable.

- (ii) [10 points] Let $X : \Omega \rightarrow \mathbb{R}$ be a random variable and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be Borel measurable. Prove that the σ -field generated by X contains the σ -field generated by $f(X)$; that is, show that

$$\sigma(f(X)) \subseteq \sigma(X).$$

Solution.

(i)

$$\Omega \xrightarrow{f} \mathbb{R}^d \xrightarrow{g} \mathbb{R}^r$$

Since f is measurable, for all $A \in \mathcal{B}(\mathbb{R}^d)$, we have $f^{-1}(A) \in \mathcal{F}$. Since g is Borel measurable, for all $B \in \mathcal{B}(\mathbb{R}^r)$, we have $g^{-1}(B) \in \mathcal{B}(\mathbb{R}^d)$.

Therefore, for all $B \in \mathcal{B}(\mathbb{R}^r)$, $g^{-1}(B) \in \mathcal{B}(\mathbb{R}^d)$, and thus $f^{-1}(g^{-1}(B)) \in \mathcal{F}$. Thus, to prove $g \circ f$ is measurable, it suffices to show that $(g \circ f)^{-1}(B) = f^{-1}(g^{-1}(B)) \in \mathcal{F}$ for all $B \in \mathcal{B}(\mathbb{R}^r)$, which we have established.

(ii)

$$\Omega \xrightarrow{X} \mathbb{R} \xrightarrow{f} \mathbb{R}$$

Since f is Borel measurable, for all $B \in \mathcal{B}(\mathbb{R})$, $f^{-1}(B) \in \mathcal{B}(\mathbb{R})$. By definition, for all $B' \in \mathcal{B}(\mathbb{R})$, $X^{-1}(B') \in \sigma(X)$. Take $B' = f^{-1}(B)$, then $X^{-1}(f^{-1}(B)) = (f \circ X)^{-1}(B) \in \sigma(X)$. Therefore, for all $B \in \mathcal{B}(\mathbb{R})$, $(f \circ X)^{-1}(B) \in \sigma(X)$, which implies $f(X)$ is measurable with respect to $\sigma(X)$, and thus $\sigma(f(X)) \subseteq \sigma(X)$.

7. [10 points] Give an example of a probability space $(\Omega, \mathcal{F}, P)$ and two random variables X and Y such that X and Y have the same distribution but are not the same random variable.

Solution.

Consider $\Omega = \{\omega_1, \omega_2\}$, $\mathcal{F} = 2^\Omega$, and $P(\{\omega_1\}) = P(\{\omega_2\}) = 1/2$. Define random variables X and Y by

$$X(\omega_1) = 0, \quad X(\omega_2) = 1, \quad Y(\omega_1) = 1, \quad Y(\omega_2) = 0.$$

Then X and Y have the same distribution, since $\mathbb{P}(X = 0) = \mathbb{P}(Y = 0) = 1/2$ and $\mathbb{P}(X = 1) = \mathbb{P}(Y = 1) = 1/2$, but X and Y are not the same random variable, since $X(\omega_1) \neq Y(\omega_1)$ and $X(\omega_2) \neq Y(\omega_2)$.